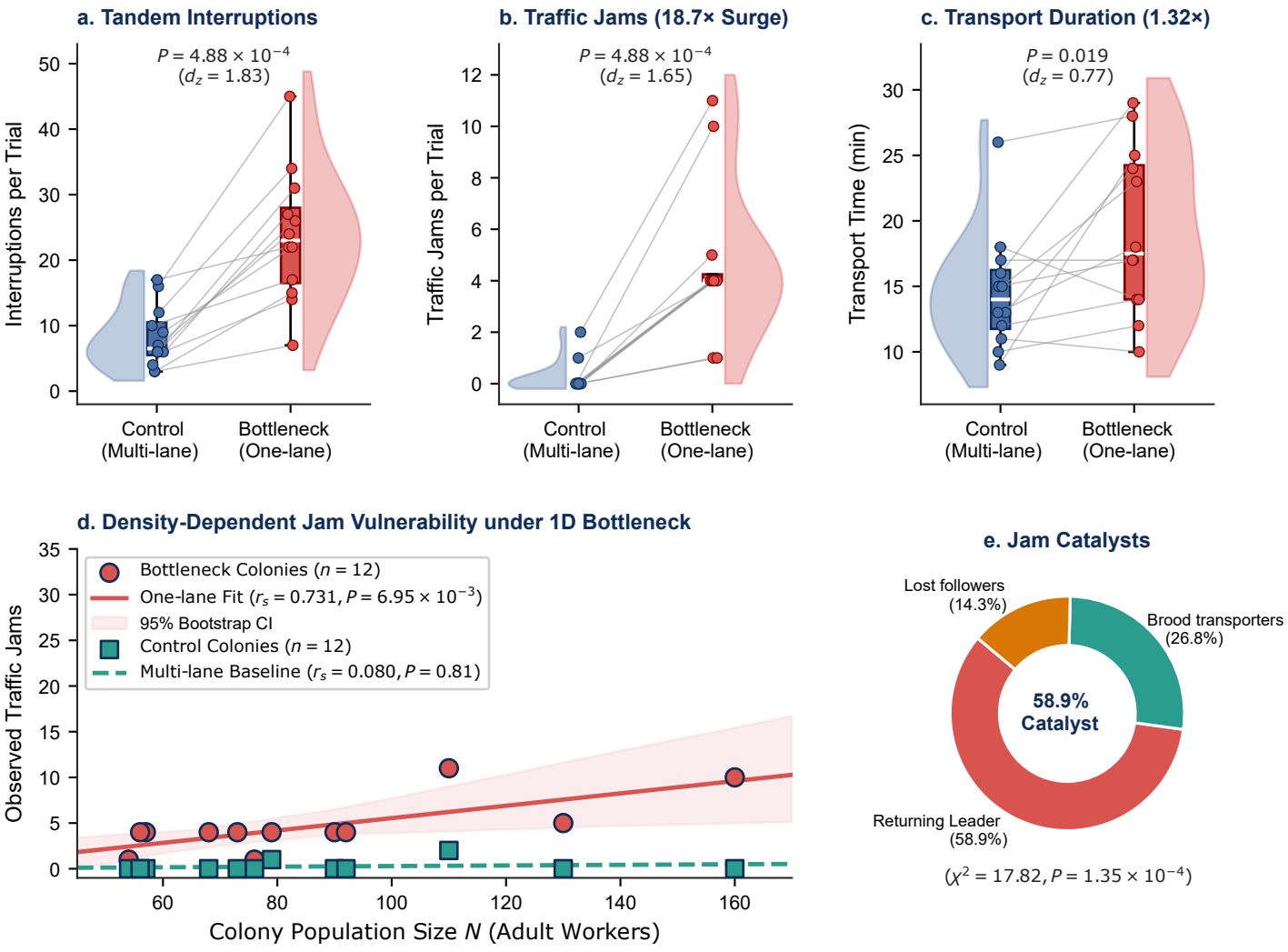

a. Tandem Interruptions
P = 4.88 × 10−4
(dz = 1.83)
Interruptions per Trial
Control (Multi-lane)
Bottleneck (One-lane)
b. Traffic Jams (18.7× Surge)
P = 4.88 × 10−4
(dz = 1.65)
Traffic Jams per Trial
Control (Multi-lane)
Bottleneck (One-lane)
c. Transport Duration (1.32×)
P = 0.019
(dz = 0.77)
Transport Time (min)
Control (Multi-lane)
Bottleneck (One-lane)
d. Density-Dependent Jam Vulnerability under 1D Bottleneck
Bottleneck Colonies (n = 12)
One-lane Fit (rs = 0.731, P = 6.95 × 10−3)
95% Bootstrap CI
Control Colonies (n = 12)
Multi-lane Baseline (rs = 0.080, P = 0.81)
Observed Traffic Jams
Colony Population Size N (Adult Workers)
e. Jam Catalysts
Lost followers (14.3%)
Brood transporters (26.8%)
58.9% Catalyst
Returning Leader (58.9%)
(χ2 = 17.82, P = 1.35 × 10−4)